

Database Design Project

Oracle Baseball League Store Database

Project Scenario:

You are a small consulting company specializing in database development. You have just been awarded the contract to develop a data model for a database application system for a small retail store called Oracle Baseball League (OBL).

The Oracle Baseball League store serves the entire surrounding community selling baseball kit. The OBL has two types of customer, there are individuals who purchase items like balls, cleats, gloves, shirts, screen printed t-shirts, and shorts. Additionally customers can represent a team when they purchase uniforms and equipment on behalf of the team.

Teams and individual customers are free to purchase any item from the inventory list, but teams get a discount on the list price depending on the number of players. When a customer places an order we record the order items for that order in our database.

OBL has a team of three sales representatives that officially only call on teams but have been known to handle individual customer complaints.

Section 4 Lesson 2 Exercise : Oracle SQL Developer Data Modeler

Engineer the Relational Model from the Logical Data Model (S4L2 Objective 3)

Part 1:

In this exercise you will take your completed logical data model that was created using Oracle SQL Data Modeler and engineer it to a Relational Model. The relational model forms the basis for the physical database design.

Create a relational model in Oracle SQL Data Modeler by following these steps:

1. Update the attributes in your entities to include their data type and size.
 - a. Double click the entity to access the properties.
 - b. Choose attributes from the left menu
 - c. Select each attribute and choose logical from the properties window to activate the data type drop down.
 - d. Choose the appropriate data type (VARCHAR to represent text, there is no VARCHAR2 option).
 - e. Place the size in the size box.
 - f. Any attributes that have been brought in as foreign keys can be renamed in the relational model by double clicking them and setting their preferred abbreviation.
 - g. Any attributes are names using keywords should have a preferred abbreviation assigned.
 - h. You cannot assign unique keys at this stage.
 - i. For any supertype entities set the subtree generation to single table in the subtype property.
2. Click the Engineer to Relational Model icon.
3. Ensure that all values are as expected by comparing the logical and relational information in tree view. Correct the values as necessary.
4. In the general option tab click engineer co-ordinates to lock the diagram in place and avoid overlap.
5. When you are happy click Engineer.
6. Expand the Relational Model node in the object browser to view the objects that you created.
7. Move the node objects to ensure that all information is visible on your diagram.
8. Check the result matches your table map and Save your work.

Task Instructions

Use the completed table mapping document produced in a previous exercise to assign the data types and sizes.

Table Name	Table Short Name			
customers	ctr			
Key Type	Optionality	Column Name	Data Type	Size
pk	*	ctr_number	VARCHAR2	6
uk	*	email	VARCHAR2	50
	*	first_name	VARCHAR2	20
	*	last_name	VARCHAR2	30
	*	phone_number	VARCHAR2	11
	*	current_balance	NUMBER	6,2
	o	loyalty_card_number	VARCHAR2	6
fk1	o	tem_id	VARCHAR2	4
fk2	o	sre_id	VARCHAR2	4

Table Name	Table Short Name			
customers_addresses	cas			
Key Type	Optionality	Column Name	Data Type	Size
pk	*	id	VARCHAR2	8
	*	address_line_1	VARCHAR2	30
	o	address_line_2	VARCHAR2	30
	*	city	VARCHAR2	15
	*	postal_code	VARCHAR2	7
fk	*	ctr_number	VARCHAR2	6

Table Name	Table Short Name			
teams	tem			
Key Type	Optionality	Column Name	Data Type	Size
pk	*	id	VARCHAR2	4
	*	name	VARCHAR2	20
	*	number_of_players	NUMBER	2
	o	discount	NUMBER	2

Table Name	Table Short Name			
sales_representatives	sre			
Key Type	Optionality	Column Name	Data Type	Size
pk	*	id	VARCHAR2	4
uk	*	email	VARCHAR2	50
	*	first_name	VARCHAR2	20
	*	last_name	VARCHAR2	30
	*	phone_number	VARCHAR2	11
	*	commission_rate	NUMBER	2
fk	*	supervisor_id	VARCHAR2	4

Table Name	Table Short Name			
sales_representatives_address	sas			
Key Type	Optionality	Column Name	Data Type	Size
Pk, fk	*	id	VARCHAR2	4
	*	address_line_1	VARCHAR2	30
	o	address_line_2	VARCHAR2	30
	*	city	VARCHAR2	15
	*	postal_code	VARCHAR2	7

Table Name	Table Short Name			
orders	odr			
Key Type	Optionality	Column Name	Data Type	Size
pk	*	id	VARCHAR2	9
	*	odr_date	DATE	
	*	odr_time	TIMESTAMP	0
	*	number_of_units	NUMBER	2
fk	*	ctr_number	VARCHAR2	6

Table Name	Table Short Name			
ordered_items	oim			
Key Type	Optionality	Column Name	Data Type	Size
	*	quantity_ordered	NUMBER	3
	*	quantity_shipped	NUMBER	3
pk, fk1	*	odr_id	VARCHAR2	9
pk, fk2	*	itm_number	VARCHAR2	10

Table Name	Table Short Name			
items	itm			
Key Type	Optionality	Column Name	Data Type	Size
pk	*	itm_number	VARCHAR2	10
	*	name	VARCHAR2	20
	*	description	VARCHAR2	50
	*	category	VARCHAR2	25
	o	color	VARCHAR2	15
	o	size	CHAR	1
fk	*	ilt_id	VARCHAR2	11

Table Name	Table Short Name			
inventory_list	ilt			
Key Type	Optionality	Column Name	Data Type	Size
pk	*	id	VARCHAR2	11
	*	cost	NUMBER	7,2
	*	units	NUMBER	4

Table Name	Table Short Name			
price_history	phy			
Key Type	Optionality	Column Name	Data Type	Size
pk	*	start_date	DATE	
pk	*	start_time	TIMESTAMP	0
	*	price	NUMBER	7,2
	o	end_date	DATE	
	o	end_time	TIMESTAMP	0
pk, fk1	*	itm_number	VARCHAR2	10

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Part 2:

In this exercise you will take your completed relational model and create the remaining constraints.

1. Identify the tables that have unique columns that have still to be created.
 - a. Right click the table and choose properties.
 - b. Select Unique Constraints from the menu.
 - c. Click the green cross to add a new unique constraint.
 - d. Use the table short name_column_name_constraint type format to name your constraints.
 - e. Select the column that is to be set as unique and click the arrow to add it to the selected column.
 - f. Click OK.

2. Identify any tables that have a recursive relationship
 - a. Right click the table and choose properties.
 - b. Select Foreign Keys from the menu.
 - c. Click the green cross to add a new foreign key.
 - d. Select the referenced column (pk) and the column that is to be used to create the link.
 - e. Click OK.

3. Check that all names satisfy naming conventions.
 - a. Go to each table and choose the Foreign Keys option in the properties dialog.
 - b. Click on the name and choose the edit icon to shorten the name
 - c. Click ok.

4. Save your work.

Task Instructions

Use the completed table mapping document produced in a previous exercise to assign the constraints.

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	*	first_name	VARCHAR2	20
	*	last_name	VARCHAR2	30
	*	phone_number	VARCHAR2	11
	*	current_balance	NUMBER	6,2
uk	o	loyalty_card_number	VARCHAR2	6
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	o	end_time	TIMESTAMP	0
pk, fk1	*	itm_number	VARCHAR2	10

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Part 3 : Create a SQL script file

In this exercise you will take your completed relational model that you created in the previous exercise and create the SQL script that will generate your tables in APEX.

Task - Creating DDL scripts

1. Ensure that you are in the relational view.
 - a. Click on the Generate DDL button on the toolbar.
 - b. Choose the environment you want to create the script for (11g is fine).
 - c. Click Generate.
 - d. In the Generation Options window double check that everything is as it should be.
 - e. Click OK.
 - f. Save your file locally as "Sports Script.ddl".
 - g. If any errors are identified then you will have to go back to your model to resolve them before regenerating the script.